

Problem: : Symmetric Radical Inequality Proo

Let a, b, c be positive reals satisfying

$$a + b + c = 3.$$

Show that

$$\sqrt{\frac{a}{b+c}} + \sqrt{\frac{b}{c+a}} + \sqrt{\frac{c}{a+b}} \leq \frac{6}{\sqrt{(a+b)(b+c)(c+a)}}.$$